

Full-precision peak representations outperform binned spectra for substructure prediction from tandem mass spectra

Xulin Pan^{1*} Chuanming Wang²

¹The Pennsylvania State University, University Park, PA, USA

²School of Biological and Food Engineering, Honghe University, Mengzi, China

Abstract

Machine-learning methods for tandem mass spectrometry (MS/MS) almost always consume the spectrum as a fixed-width binned intensity vector, a representation inherited from spectral library search. Binning is rarely treated as a modelling decision and its cost has not been measured. Here we price it directly. Using the CASMI 2026 corpus we trained two models that differ only in how the spectrum enters: a 0.5 Da binned vector, and a set of peaks whose m/z is retained at full precision and embedded with an explicit mass-defect channel. Data, structural targets, optimiser, training budget and evaluation were held fixed, and the binned model was given 63% more parameters. On 8,692 held-out molecules the peak-set representation raised macro AUPRC from 0.0476 to 0.1119 against a chance level of 0.0093, a paired per-target gain of +0.0644 (95% CI +0.0570 to +0.0722) that improved 89.7% of targets individually. The representation change was more than twice the size of an architectural change (hierarchical multi-view pooling) measured on the same data. Binning at 0.5 Da absorbs 37.5% of the test peaks a model actually receives into bins already occupied by another peak, spans 1,667 ppm at m/z 300 on an instrument delivering about 5 ppm, and pools hundreds of candidate fragment compositions into one coordinate. We conclude that a binned baseline is no longer an adequate control for MS/MS structure annotation. Code and configurations are available at <https://github.com/xulinpan/cheminformatics>.

Scientific contribution

We provide the first controlled price for the spectral input representation in MS/MS substructure prediction, isolating it from architecture, capacity and training budget, and granting the weaker representation the larger parameter count. The measured effect exceeds that of the architectural component built on top of it by a factor of two, and we quantify the mechanism by which binning destroys information: 37.5% of the peaks a model receives are absorbed into shared bins, discarding the mass defect that distinguishes isobaric fragment compositions. The practical consequence is a change in what counts as an adequate control in this field.

Keywords

Tandem mass spectrometry, Structure annotation, Spectral representation, Mass defect, Substructure prediction, Metabolite identification

*Correspondence: xxp5066@psu.edu

Introduction

Assigning a chemical structure to an MS/MS spectrum is the rate-limiting step in untargeted metabolomics. Where the molecule is present in a reference library the problem reduces to spectral matching, but most spectra acquired from biological extracts are not [1]. For the remainder, computational annotation proceeds by relating the spectrum to candidate structures: by in-silico fragmentation and scoring, as in MetFrag [2]; by predicting a molecular fingerprint and searching a structure database, as in CSI:FingerID [3] and MIST [4]; or by learning a spectrum representation whose geometry reflects chemical similarity, as in MS2DeepScore [5] and DreaMS [6].

Nearly all of these systems begin from the same input object. The spectrum, which is physically a sparse list of (m/z , intensity) pairs, is discretised onto a fixed grid of bins, typically 0.01 to 1 Da wide, and handed to the model as a dense vector. The convention is so widespread that it is seldom stated as a choice.

Its origin lies in library search. Cosine similarity between two binned vectors is inexpensive, and a bin comfortably wider than the instrument’s mass error makes matching robust to calibration drift between two laboratories. Those are appropriate properties for comparing two measurements of the same compound. They are not obviously appropriate for inferring an unknown structure, and the two problems have been treated as though a single representation serves both. A 0.5 Da bin spans roughly 1,700 ppm at m/z 300 while a modern time-of-flight instrument delivers on the order of 5 ppm; everything the instrument resolves within that window is discarded. What is discarded is not noise. The mass defect, the difference between a fragment’s exact mass and its nominal mass, is the quantity that distinguishes C_5H_7^+ (67.0542) from $\text{C}_4\text{H}_3\text{O}^+$ (67.0178), and therefore the quantity that carries elemental composition.

Parts of the field have moved away from naive binning. CSI:FingerID scores against fragmentation trees built from exact masses, MIST tokenises peaks by chemical formula, DreaMS operates on peak lists, and TransExION explicitly separates nominal mass from an embedded mass defect when aligning fragments between spectra [7]. These are strong systems, and each differs from a binned baseline in architecture, training data and objective simultaneously. None of them isolates the representation, so the field has no estimate of what the choice is worth on its own, and binned baselines continue to appear as controls in new work.

This paper supplies that estimate. We train two models that differ in exactly one respect — whether the spectrum is discretised — while holding the corpus, structural targets, splits, optimiser, schedule and evaluation fixed, and we give the binned model the larger parameter budget so that any deficit cannot be attributed to capacity. We report the effect size with paired intervals over targets, quantify the information binning removes, and compare the result against an architectural change measured under the same protocol.

Methods

Corpus and splits

We used the CASMI 2026 training corpus, an aggregation of eleven public and in-house MS/MS libraries. All experiments here use a fixed subset of two of its twenty-one Parquet row groups: 262,041 spectra over 43,536 distinct structures, comprising 15,299,995 peaks after cleaning. Spectra were retained if the precursor m/z was below 2,000 and the reported precursor error was within 20 ppm; peaks were retained above 0.1% of the base peak and below the precursor plus 1.5 Da, capped at

the 256 most intense, and m/z values were kept in double precision throughout preprocessing.

Structures were assigned to five folds by grouping generic Murcko scaffolds, with acyclic structures falling back to grouping by their own identifier. Fold sizes were balanced greedily (8,707 to 8,708 molecules). Folds 2–4 were used for fitting, fold 1 for model selection, and fold 0 was held out; all reported numbers are on fold 0 (8,692 molecules).

Structural targets

Each structure was decomposed with BRICS and represented by the presence or absence of each of the 512 most frequent attachment-labelled fragments, the vocabulary being fitted on the training folds alone. The resulting target matrix is 97.1% zero, with a median of 17 positive targets per molecule. On the held-out fold 477 targets carry at least five positives, at a mean prevalence of 0.0093; that prevalence is the chance level for macro AUPRC. Because 475 of 600 candidate targets fall below 1% prevalence, we report AUPRC as the primary metric and AUROC only as a secondary summary.

Representation A: binned vector (control)

Each spectrum was projected onto 2,048 fragment bins and 2,048 precursor-minus-fragment bins of width 0.5 Da over $[0, 1024)$, with square-root intensity accumulated into each bin and the concatenated vector L_2 -normalised. The 4,096-dimensional vector was encoded by a two-layer perceptron. This is the representation used, with minor variation in bin width, throughout the MS/MS machine-learning literature.

Representation B: full-precision peak set

Each peak was encoded as a token without discretisation. For a peak at m/z m in a spectrum with precursor m^{prec} we formed a multi-scale sinusoidal embedding

$$\varphi_L(m) = [\sin(2\pi m/\lambda_1), \cos(2\pi m/\lambda_1), \dots, \sin(2\pi m/\lambda_L), \cos(2\pi m/\lambda_L)]^\top, \quad (1)$$

with $L = 32$ wavelengths spaced geometrically from 2 mDa to 10^3 Da, applied both to m and to the neutral loss $m^{\text{prec}} - m$. Short wavelengths resolve the mass defect and long wavelengths carry nominal mass. The token also received the explicit defect $\delta = m - \text{round}(m)$, square-root and log-transformed intensity, and intensity rank.

One numerical detail is essential rather than incidental. At a wavelength of 2 mDa and m/z 342 the phase in Eq. (1) exceeds 10^6 , and evaluating it in single precision destroys exactly the low-order information the embedding exists to preserve. We therefore compute the phase in double precision and reduce it modulo 2π before casting. With this reduction the embeddings of $\text{C}_4\text{H}_3\text{O}^+$ and C_5H_7^+ , which share a 0.5 Da bin, differ by 1.82 in the infinity norm, and two peaks 1 mDa apart at m/z 342 remain distinguishable.

Tokens were encoded by a four-block Transformer over the peak set with a learned pooling token, each block conditioned on the acquisition covariates (adduct, polarity, instrument family, harmonised collision energy) by adaptive layer normalisation. The encoder is permutation invariant and accepts variable peak counts.

Training protocol

Both models were trained for fifteen epochs with the AdamW optimiser, a one-cycle schedule with peak learning rate 10^{-3} , batches of 16 molecules, mixed precision, and gradient clipping at 1.0. The loss was binary cross-entropy over the 512 presence targets, with an auxiliary negative-binomial term on 88 functional-group counts. Checkpoints were selected on validation macro AUPRC. No

hyperparameter search was performed for either arm. The binned model has 1,164,560 parameters and the peak-set model 712,336: the control carries 63% more capacity than the treatment.

Evaluation

We report macro AUPRC over targets with at least five held-out positives, together with macro AUROC, micro AUPRC and Brier score. Because the two arms are evaluated on identical molecules and identical targets, the appropriate contrast is paired: for each target we compute the difference in average precision and bootstrap the mean of those differences over targets (5,000 resamples). We additionally report the fraction of targets improved, which is insensitive to a few large per-target swings.

Results

What binning removes

We first quantified the information discarded before any model is fitted (Fig. 1). Across the 93,611 peaks of the evaluation spectra that survive preprocessing and are therefore seen by both arms, 35,116 — 37.5% — fall into a 0.5 Da bin already occupied by another peak from the same spectrum and are therefore absorbed. The loss is not concentrated in any mass range, running between 29% and 41% across the observed m/z window. The observed mass defect spans $[-0.47, +0.47]$ Da and shows clear compositional structure, most visibly the discrete combs below m/z 150 where few elemental compositions are available. Enumerating plausible CHNOS compositions within 10 ppm places 328 candidate fragment formulas inside a single 0.5 Da bin at m/z 300.

The denominator matters here. The raw test files contain 400,707 peaks, of which 60.7% collide under 0.5 Da binning, but preprocessing removes 76.6% of those as sub-threshold before either arm is trained. Charging binning for peaks no model ever receives would overstate the loss, so we report the absorbed fraction over the cleaned spectra throughout.

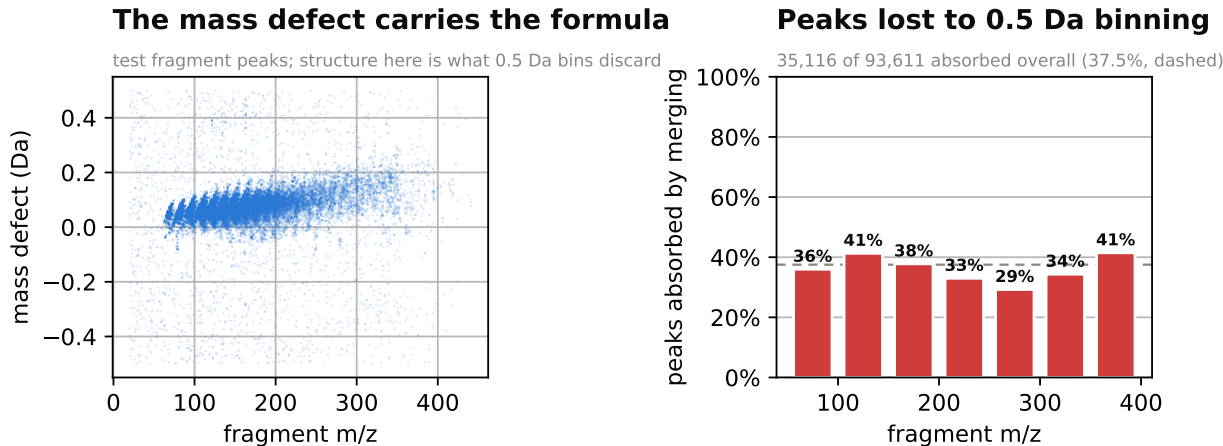


Figure 1: Information discarded by 0.5 Da binning. **Left:** mass defect against fragment m/z for the evaluation spectra; the visible structure, including the discrete combs at low m/z , is what binning removes. **Right:** fraction of peaks absorbed into an already-occupied bin, by mass range; the dashed line marks the overall 37.5%.

Controlled comparison

Replacing the binned vector with the full-precision peak set raised macro AUPRC from 0.0476 to 0.1119, from 5.1 to 12.0 times the chance level of 0.0093 (Table 1, Fig. 2). Macro AUROC rose from 0.7832 to 0.8734, micro AUPRC from 0.2905 to 0.3842, and the Brier score improved from 0.00704 to 0.00653. The deficit cannot be attributed to capacity, since the binned model holds the larger parameter count.

Table 1: Ablation on 8,692 held-out molecules. Chance macro AUPRC is 0.0093 over 477 scored targets.

rung	encoder	pooling	parameters	macro AUPRC	macro AUROC	micro AUPRC	Brier
M0	0.5 Da bins	mean	1,164,560	0.0476	0.7832	0.2905	0.00704
M1	peak set	mean	712,336	0.1119	0.8734	0.3842	0.00653
M2	peak set	hierarchical	829,584	0.1460	0.8861	0.4088	0.00639

The paired per-target contrast gives +0.0644 average precision with a 95% bootstrap interval of +0.0570 to +0.0722, and 89.7% of individual targets improve (Table 2). The interval excludes zero by a wide margin and the improvement is broad rather than driven by a small number of targets.

Table 2: Paired per-target AUPRC differences, bootstrapped over targets.

comparison	change	Δ AUPRC	95% CI	targets improved
M1 – M0	representation	+0.0644	[+0.0570, +0.0722]	89.7%
M2 – M1	aggregation	+0.0341	[+0.0277, +0.0407]	72.3%
M2 – M0	both	+0.0984	[+0.0885, +0.1090]	94.3%

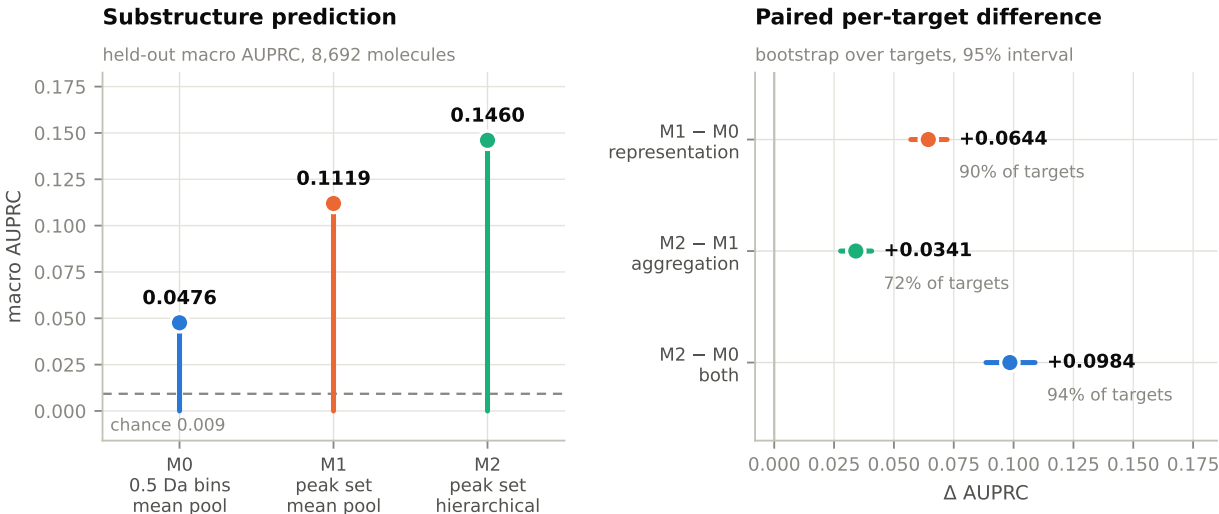


Figure 2: Controlled comparison on 8,692 held-out molecules. **Left:** macro AUPRC for the binned control (M0), the peak-set model (M1) and a peak-set model with hierarchical multi-view pooling (M2), against the chance level. **Right:** paired per-target differences with 95% bootstrap intervals over targets.

The representation effect exceeds the architectural effect

To place the result on a scale, we measured an architectural change under the same protocol:

replacing mean pooling over a molecule’s spectra with a hierarchical measurement model that weights each spectrum by a learned, acquisition-dependent precision. That change is worth +0.0341 average precision (95% CI +0.0277 to +0.0407), improving 72.3% of targets. The representation change is therefore 1.9 times larger than the architectural change built on top of it, measured on the same molecules with the same targets.

The comparison is not confounded by convergence

Both arms plateaued well inside the training budget (Fig. 3). Validation macro AUPRC for the binned model moved from 0.04504 at epoch 12 to 0.04540 at epoch 14, and for the peak-set model from 0.10645 to 0.10670 over the same interval. Neither arm is undertrained relative to the other, and extending the schedule would not change the ordering.

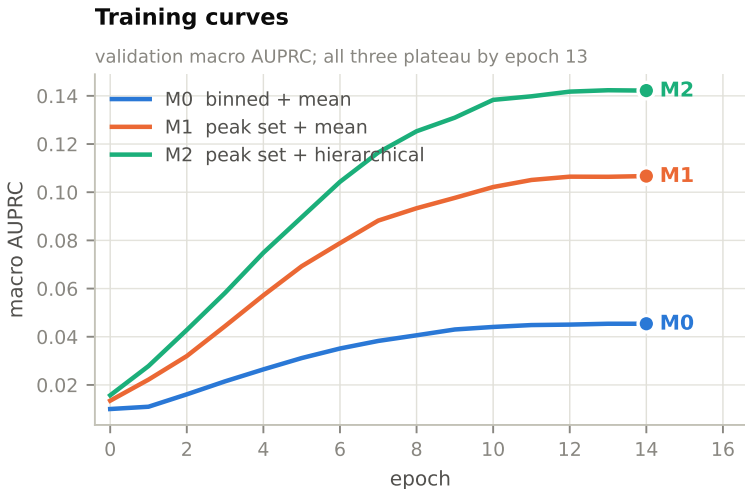


Figure 3: Validation macro AUPRC by epoch. Both arms plateau by epoch 13, so the comparison does not depend on unequal convergence.

Discussion

The result is a single number with a narrow interval, and its interest lies in what it is a number *about*. Discretising the m/z axis is not a preprocessing detail on the way to a model; on this corpus it costs more than the model.

The reason binning persists is worth stating plainly, because it is a good reason that has outlived its context. In library search a wide bin buys robustness: two laboratories measuring the same compound will not agree to the fourth decimal place, and a bin wider than their disagreement makes the match succeed. That argument is about tolerance between two measurements. It says nothing about how much information a single measurement carries, and structure annotation is an inference problem rather than a matching problem. When the same representation was carried across, the tolerance argument came with it and the information argument was never made.

Our finding is consistent with the direction the strongest systems have taken independently. CSI:FingerID scores against fragmentation trees over exact masses [3], MIST tokenises by chemical formula [4], DreaMS learns from peak lists [6], and TransExION embeds the mass defect as a channel distinct from nominal mass [7]. Each of those systems differs from a binned baseline in

several respects at once, so none of them prices the representation alone; what we add is the controlled estimate, and the observation that it is large.

The practical consequence is a claim about controls. A new method evaluated against a binned baseline inherits a handicap in its comparator that may exceed the effect it reports. On the evidence here, a difference of the size typically claimed for an architectural contribution can be produced by changing the input representation and nothing else. We would suggest that a binned baseline is no longer sufficient, and that methods papers in this area report their comparator’s representation as explicitly as its architecture.

Limitations

Three limits bound the claim. The experiments use approximately a tenth of the available corpus; while we do not expect the ordering to invert, absolute values will move with scale. Each arm was trained once, and the reported intervals are bootstraps over targets, which capture target-to-target variation but not initialisation or run-to-run variance; replication across seeds is required before the effect size is treated as settled. And the comparison uses one target family (BRICS substructure presence), one instrument family and one bin width; a narrower bin would recover part of the loss, and quantifying the effect as a function of bin width would sharpen the recommendation. We report the result as a lower bound on the cost of a widely used bin width rather than as a characterisation of binning in general.

Conclusion

Binned intensity vectors entered MS/MS machine learning from spectral library search, where a bin wider than the instrument’s error is a virtue. Carried into structure annotation, the same choice discards 37.5% of the peaks a model receives and with them the mass defect that identifies fragment compositions. In a controlled comparison holding data, targets, optimiser and schedule fixed, and giving the binned arm 63% more parameters, replacing the binned vector with a full-precision peak set raised macro AUPRC from 0.0476 to 0.1119 — a paired gain of +0.0644 that improved 89.7% of targets and exceeded the architectural change measured on the same data by a factor of 1.9. We conclude that the input representation should be treated as a modelling decision with a measurable cost, and that a binned baseline is no longer an adequate control for this task.

Supplementary Information

Additional file 1 reports per-target average precision for both arms, the full bootstrap distribution of the paired difference, and the peak-absorption analysis broken down by adduct and collision energy.

Acknowledgements

We thank the CASMI 2026 organisers for assembling and releasing the corpus.

Author contributions

X.P. designed the study, implemented the models, performed the experiments and analyses, and wrote the manuscript. C.W. contributed to the interpretation of the chemical results and revised the manuscript. Both authors read and approved the final manuscript.

Funding

Not applicable.

Availability of data and materials

The CASMI 2026 corpus is distributed by the competition organisers. All code, configurations and the exact preprocessing used here are available at <https://github.com/xulinpan/cheminformatics>. Figures and tables are regenerated from the saved result files by a single command.

Declarations

Competing interests

The author declares no competing interests.

References

- [1] Hoffmann MA, Nothias L-F, Ludwig M, Fleischauer M, Gentry EC, Witting M, Dorrestein PC, Dührkop K, Böcker S (2022) High-confidence structural annotation of metabolites absent from spectral libraries. *Nat Biotechnol* 40:411–421
- [2] Ruttkies C, Schymanski EL, Wolf S, Hollender J, Neumann S (2016) MetFrag relaunched: incorporating strategies beyond in silico fragmentation. *J Cheminform* 8:3
- [3] Dührkop K, Shen H, Meusel M, Rousu J, Böcker S (2015) Searching molecular structure databases with tandem mass spectra using CSI:FingerID. *Proc Natl Acad Sci USA* 112:12580–12585
- [4] Goldman S, Wohllwend J, Stražar M, Haroush G, Xavier RJ, Coley CW (2023) Annotating metabolite mass spectra with domain-inspired chemical formula transformers. *Nat Mach Intell* 5:965–979
- [5] Huber F, van der Burg S, van der Hooft JJJ, Ridder L (2021) MS2DeepScore: a novel deep learning similarity measure to compare tandem mass spectra. *J Cheminform* 13:84
- [6] Bushuiev R, Bushuiev A, Samusevich R, Brungs C, Sivic J, Pluskal T (2025) Self-supervised learning of molecular representations from millions of tandem mass spectra using DreaMS. *Nat Biotechnol*
- [7] Bui-Thi D, Liu Y, Lippens JL, Laukens K, De Vijlder T (2024) TransExION: a transformer based explainable similarity metric for comparing IONS in tandem mass spectrometry. *J Cheminform* 16:61